

Title : Case study on Amazon EC2 and learn about Amazon EC2 web services.

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Roll No. : - CO3110

Class : - TE Computer

Subject : - Cloud

Computing

Introduction:

Amazon Web Services is the world's most comprehensive and broadly adopted cloud, enabling customers to build almost anything they can imagine. They offer the greatest choice of innovative cloud and AI capabilities and expertise, on the most extensive global infrastructure, with industry-leading security, reliability, and performance.

Benefits and Features:

1. SLA commitment

Access reliable, scalable infrastructure on demand. Scale capacity within minutes with SLA commitment of 99.99% availability.

2. AWS Nitro System

Provide secure computers for your applications. Security is built into the foundation of Amazon EC2 with the AWS Nitro System.

3. Optimize performance and cost

Optimize performance and cost with flexible options like AWS Graviton-based instances, Amazon EC2 Spot instances, and AWS Savings Plans.

4. AWS Migration Tools

Migrate and build apps with ease using AWS Migration Tools, AWS Managed Services, or Amazon Lightsail. Learn how AWS can help.

Use Cases

1. Run cloud-native and enterprise applications

Amazon EC2 delivers secure, reliable, high-performance, and cost-effective compute infrastructure to meet demanding business needs.

2. Develop for Apple platforms

Build, test, and sign on-demand macOS workloads. Access environments in minutes, dynamically scale capacity as needed, and benefit from pay-as-you-go pricing.

3. Scale for HPC applications

Access the on-demand infrastructure and capacity you need to run HPC applications faster and cost-effectively.

4. Train and deploy ML applications

Amazon EC2 delivers the broadest choice of compute, networking (up to 400 Gbps), and storage services purpose-built to optimize price performance for ML projects.

Related services

1.Lightsail

This option is best if you want a completely managed infrastructure for simple web applications, websites, and development environments at a fixed monthly pricing. Find out more about Lightsail compared to EC2.

2.Elastic Container Service

This option is best if you want a completely managed container orchestration platform with global scalability.

3.Lambda

This option is best if you want to run code without thinking about servers.

4.Elastic Beanstalk

This option is best if you want to write and deploy code for web applications with JavaScript or write your own code in Java, .NET, Go, Ruby, PHP, Node.js, or Python.

Instance

EC2 is an on-demand computing service on the AWS cloud platform called instances. It lets you rent virtual computers to run your applications. You pay only for what you use.

Instance Types

Instance types are the hardware profiles of your virtual server. AWS offers a vast array of instance types optimized for different tasks, grouped into families:

- General purpose

General purpose instances provide a balance of compute, memory and networking resources, and can be used for many workloads. These instances are good for applications such as web servers, code repositories, and small-to-medium databases.

- Compute optimized

Compute optimized instances are ideal for compute bound applications that benefit from high-performance processors. Some examples of workloads for compute instances are batch processing, media transcoding, and dedicated game servers.

- Memory optimized

Memory optimized instances are designed to deliver fast performance for workloads that process large data sets in memory. For example, these instances are good for in-memory databases, data analytics, and enterprise applications.

- Accelerated computing

Accelerated computing instances use hardware accelerators, or co-processors, to perform functions more efficiently. For example, they can perform floating point number calculations, graphics processing, or data pattern matching.

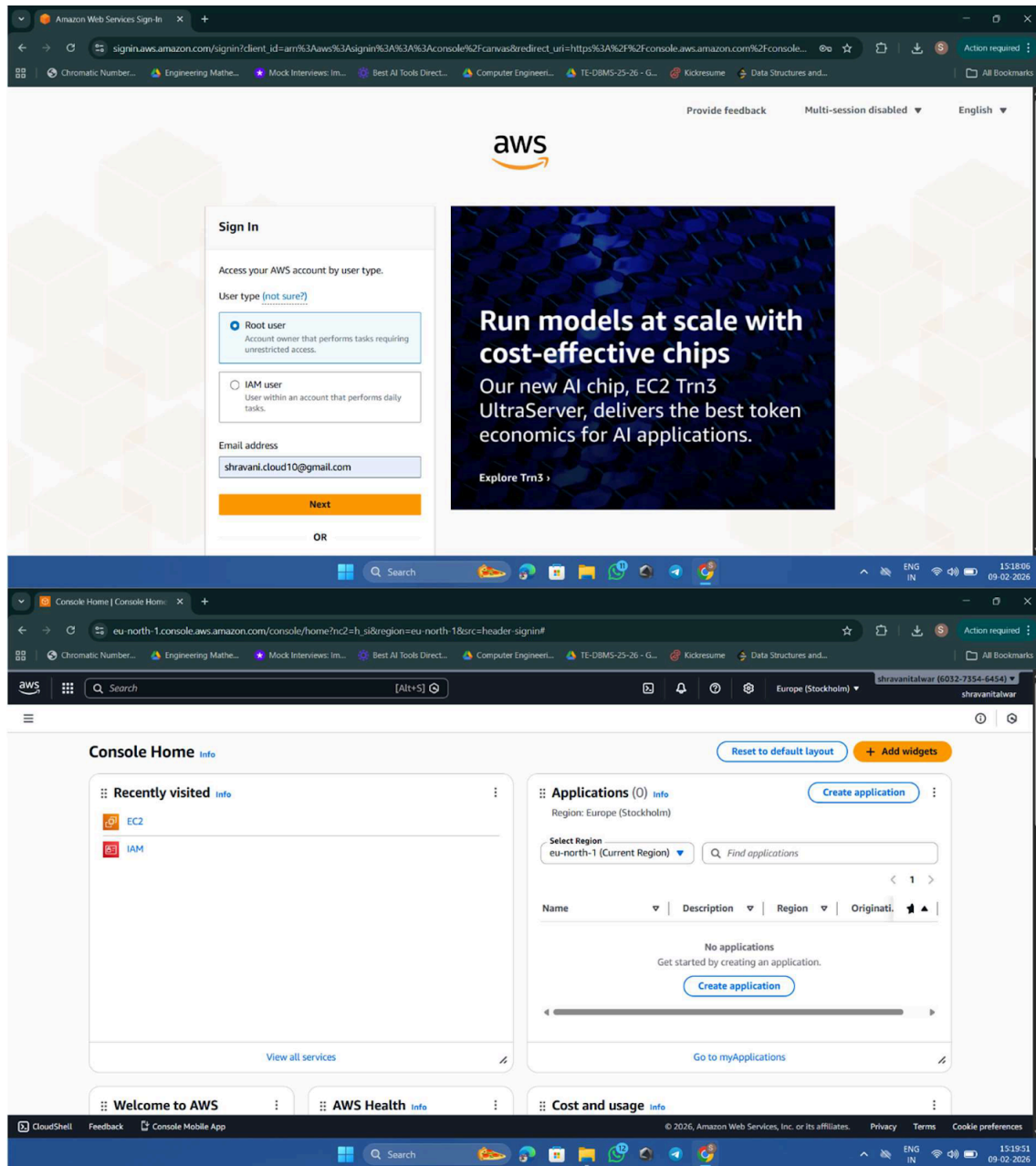
- Storage optimized

Storage optimized instances deliver millions of low-latency, random I/O operations per second to applications. They're designed for workloads that require high, sequential read and write access to very large data sets on local storage. For example, they're good for high-throughput databases, data processing, and data

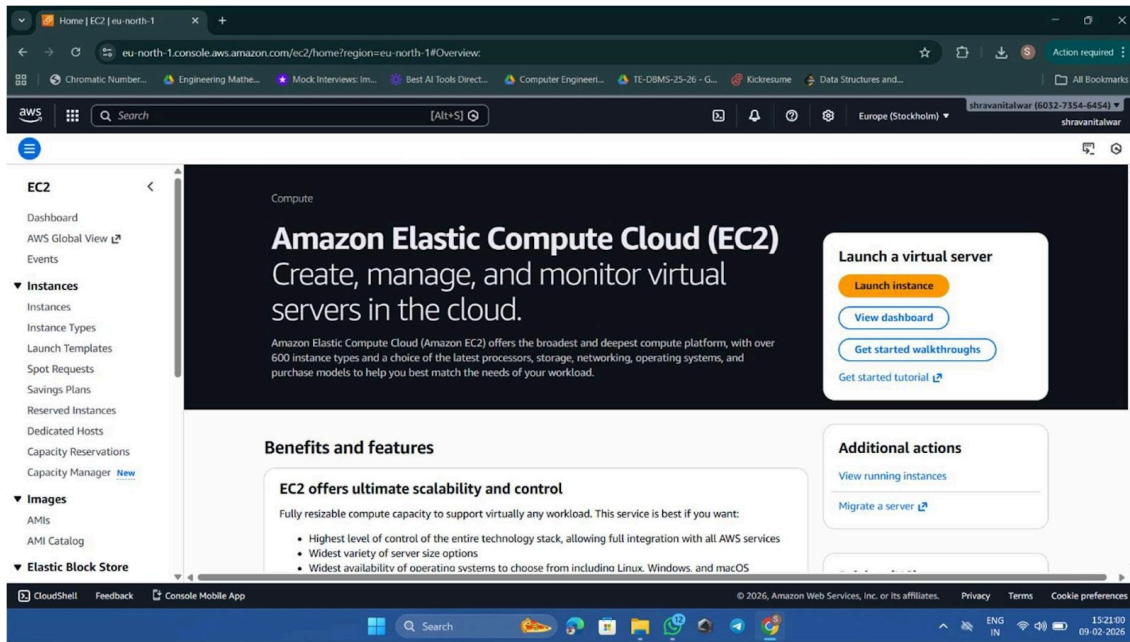
- HPC optimized

High-performance computing (HPC) instances offer the best price performance for running HPC workloads at scale. HPC instances are ideal for applications that benefit from high-performance processors such as complex simulations, deep learning, and visual effects rendering.

Step 1 : Sign-in to AWS account and open IAM console by using the following link - <https://console.aws.amazon.com/iam/> .

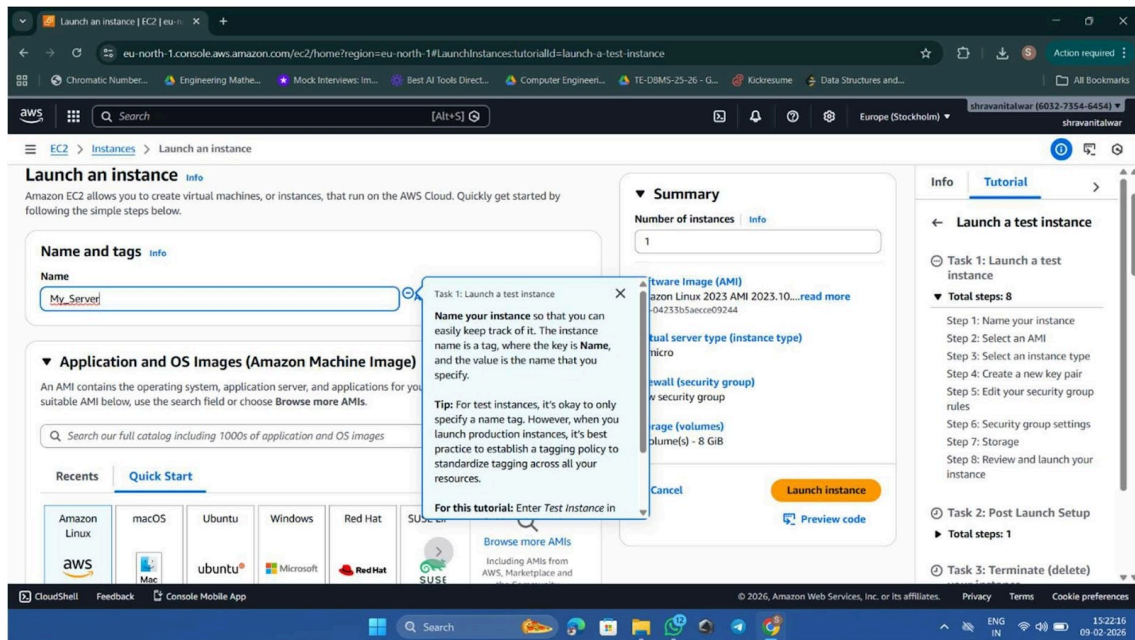


Step 2 : Navigate to EC2 instance page

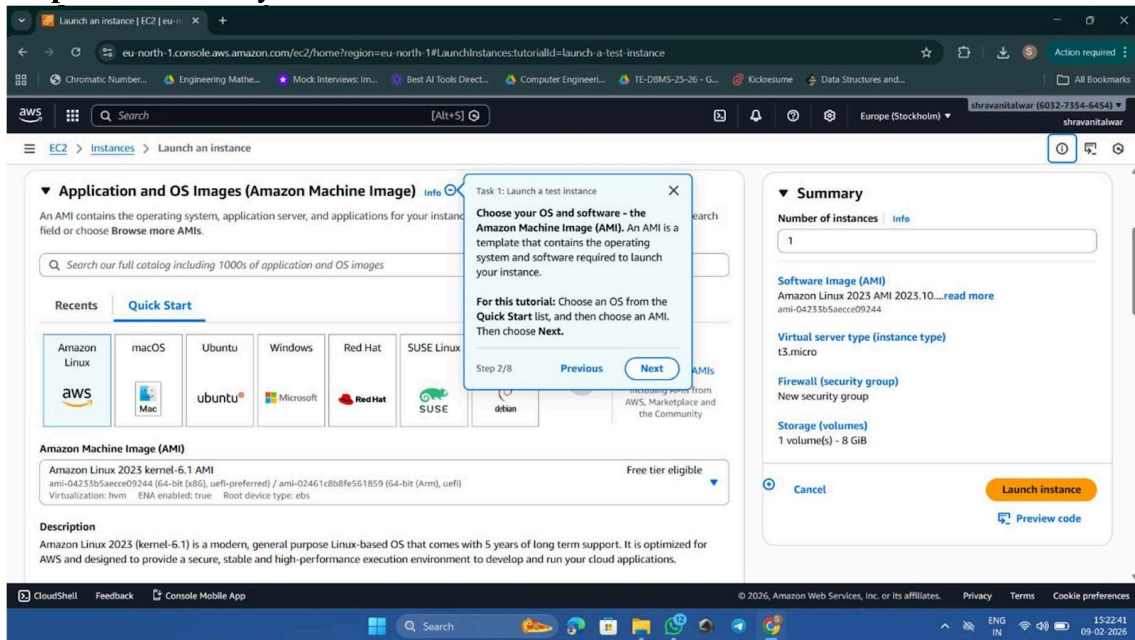


Step 3 : Follow following steps to create EC2 instance in VM using required AMI

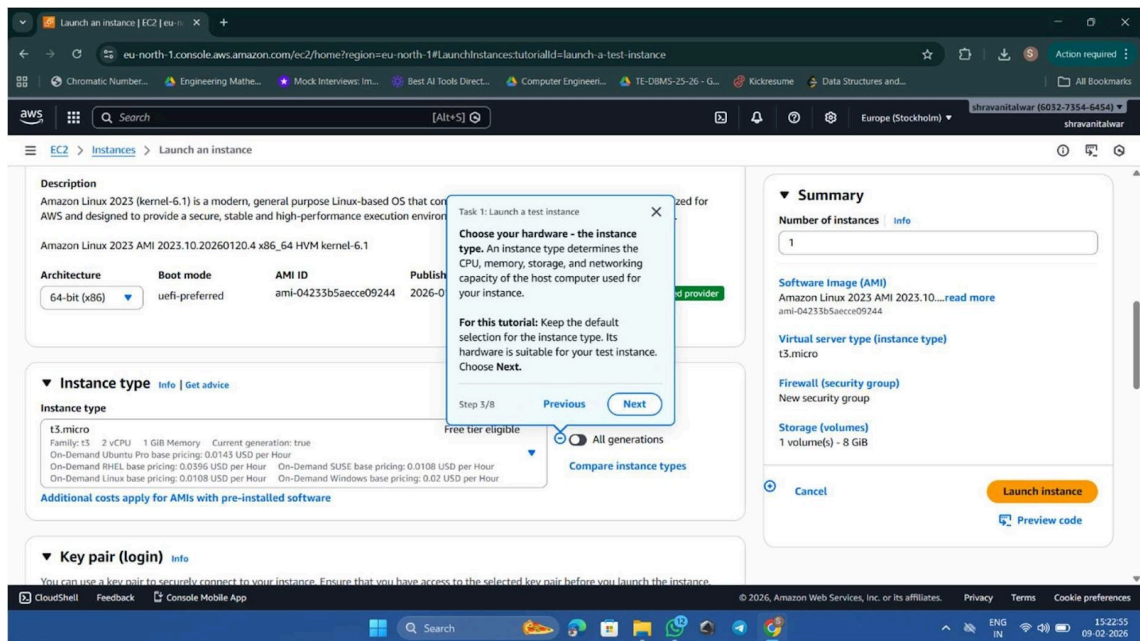
Step 3a : Name an instance



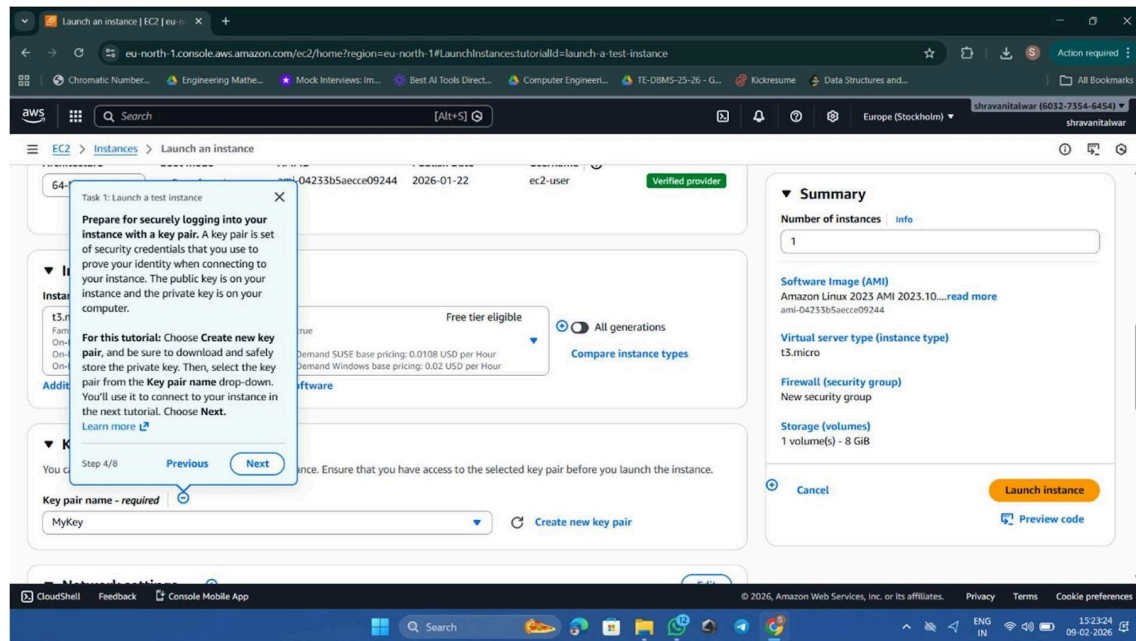
Step 3b : Choose your OS and Software i.e. AMI



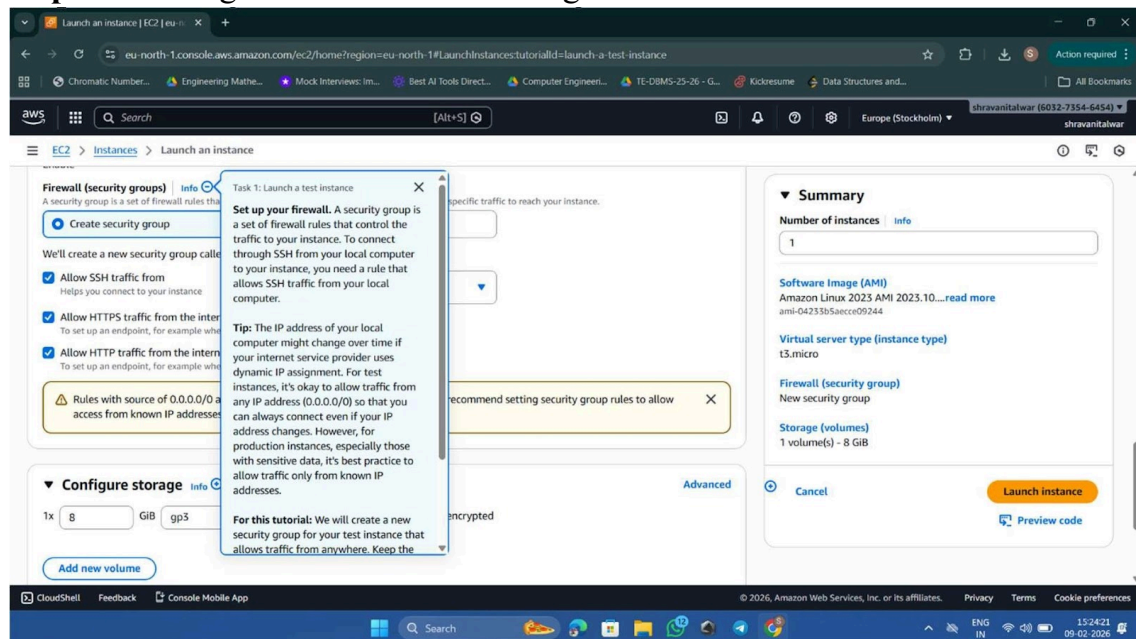
Step 3c : Choose your hardware i.e. instance type



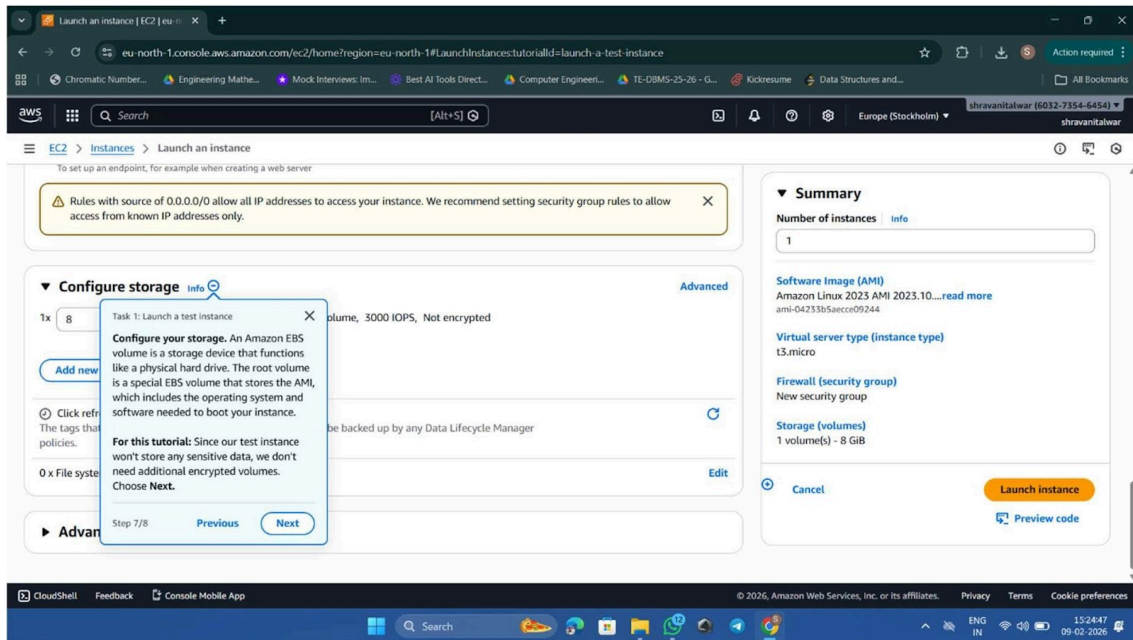
Step 3d : Create Key pair for authorization



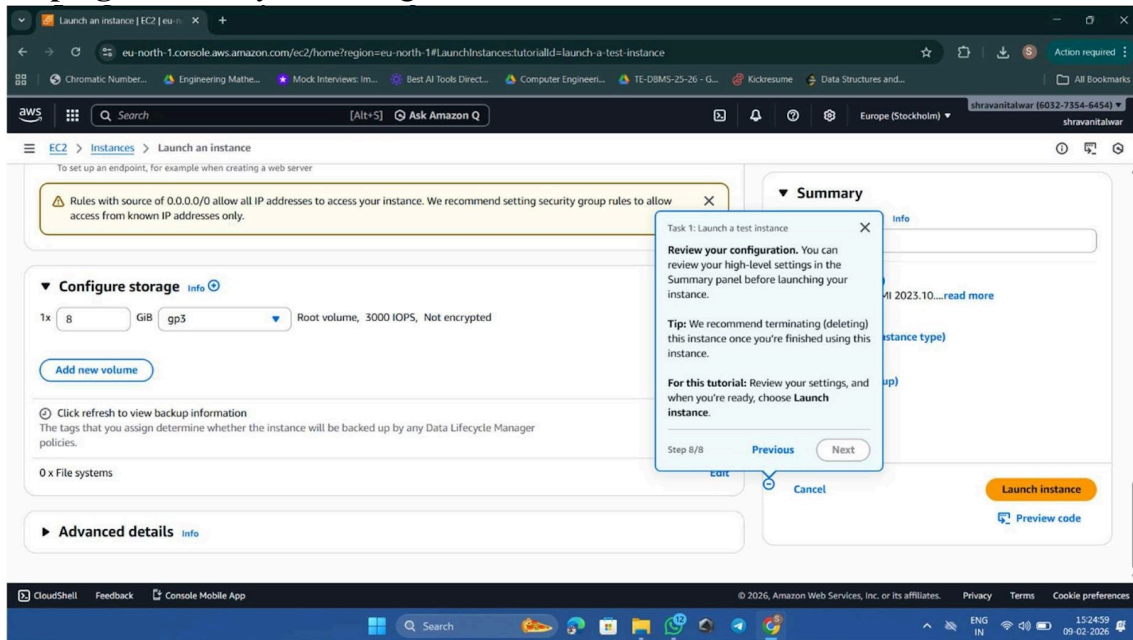
Step 3e : Configure network for enabling network access



Step 3f : Configure your storage



Step 3g : Review your configuration and Launch the instance



Step 4 : Connect to the instance and Start Working

The image shows two screenshots of the AWS Management Console. The top screenshot displays the 'Launch an instance' page for instance `i-086e5889de81b8726`. A green success banner at the top states: 'Successfully initiated launch of instance (i-086e5889de81b8726)'. Below this, the 'Launch log' section is visible. The 'Next Steps' section provides guidance on what to do next. A 'Task 1: Post launch setup' pop-up window is overlaid, explaining that once the instance is up and running, users can connect to it and click the 'connect to instance' button. The 'Next Steps' section includes three main actions: 'Create billing usage alerts', 'Connect to instance' (which is highlighted with a blue border and a 'Next' button), and 'Connect an RDS database'. The bottom screenshot shows the 'Connect to instance' page for the same instance. It offers two connection methods: 'Connect using a Public IP' (selected) and 'Connect using a Private IP'. Under 'Public IP', the 'Public IPv4 address' is listed as `16.16.167.184`. The 'Username' field is set to `ec2-user`. A note at the bottom states: 'In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.' The 'Connect' button is highlighted in orange.

Launch an instance | EC2 | eu-north-1

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#LaunchInstances:tutorialId=launch-a-test-instance

Success
Successfully initiated launch of instance (i-086e5889de81b8726)

Launch log

Next Steps

What would you like to do next with this instance, for example?

Create billing usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.
[Create billing alerts](#)

Connect to instance
Once your instance is up and running, you can connect (log into) it. It may take a few minutes after clicking the launch instance button for your instance to begin running. Once your instance is running, click the 'connect to instance' button.
[Learn more](#)
Step 2/2 [Previous](#) [Next](#)
[Connect to instance](#)
[Learn more](#)

Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
[Connect an RDS database](#)
[Create a new RDS database](#)
[Learn more](#)

Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots.
[Create EBS snapshot policy](#)

Task 1: Post launch setup

Once your instance is up and running, you can connect (log into) it. It may take a few minutes after clicking the launch instance button for your instance to begin running. Once your instance is running, click the 'connect to instance' button.

CloudShell Feedback Console Mobile App

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Launch an instance | EC2 | eu-north-1

Connect to instance | EC2 | eu-north-1

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#ConnectToInstance:instanceId=i-086e5889de81b8726

EC2 Instance Connect Session Manager SSH client EC2 serial console

Instance ID
i-086e5889de81b8726 (My_Server)

Connection type

☒ Connect using a Public IP
Connect using a public IPv4 or IPv6 address

☐ Connect using a Private IP
Connect using a private IP address and a VPC endpoint

Public IPv4 address
16.16.167.184

IPv6 address
-

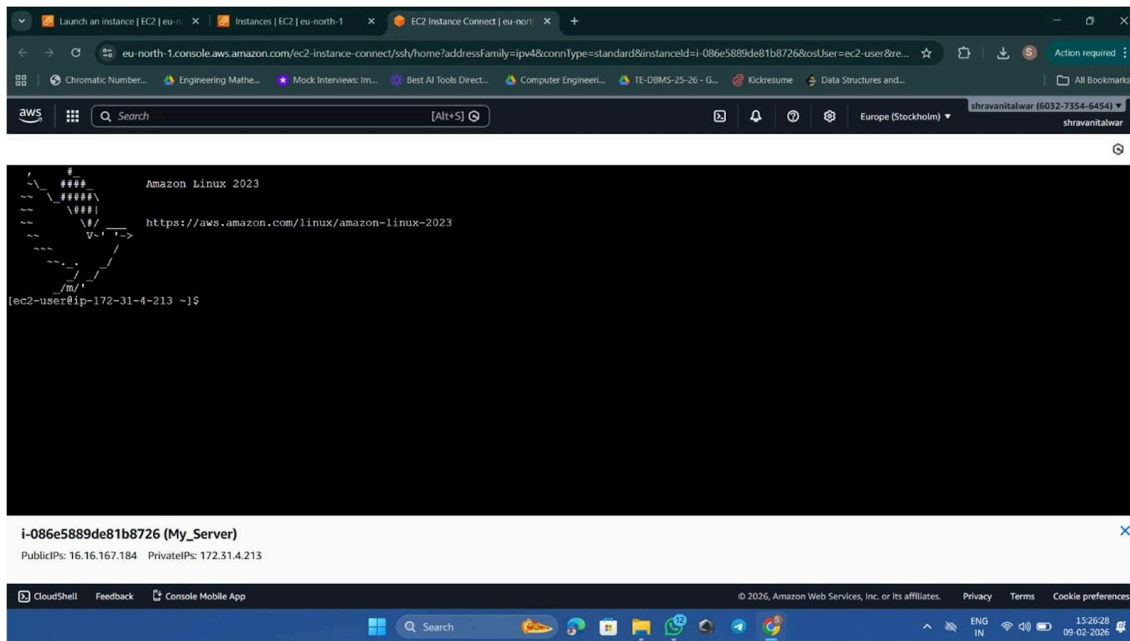
Username
Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2-user.
ec2-user

Note: In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel Connect

CloudShell Feedback Console Mobile App

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Launch Templates

Use launch templates to automate instance launches, simplify permission policies, and enforce best practices across your organization. Save launch parameters in a template that can be used for on-demand launches and with managed services, including EC2 Auto Scaling and EC2 Fleet. Easily update your launch parameters by creating a new launch template version.

Benefits:

Streamline provisioning

Simplify permissions

Governance

Spot Savings:

Spot savings are estimated savings and may differ from actual savings. This is because the savings shown on this page do not include the billing adjustments for your usage.

Savings Plans:

Savings Plans allow you to easily reduce your bill by making a commitment to compute usage (e.g. \$10/hour) instead of making commitments to specific instance configurations or compute services. Amazon Web Services offers two types of Savings Plans - Compute Savings Plans and EC2 Instance Savings Plans.

Reserved Instances

The Reserved Instances page lists all of your Reserved Instances in the currently selected AWS Region.

Dedicated Hosts

Amazon EC2 Dedicated Hosts allow you to use your eligible software licenses from vendors such as Microsoft and Oracle on Amazon EC2, so that you get the flexibility and cost effectiveness of using your own licenses, but with the resiliency, simplicity and elasticity of AWS.

Capacity Reservations

Capacity Reservations allow customers to reserve compute capacity for EC2 instances in a specific Availability Zone.

Capacity Manager

Capacity Manager aggregates EC2 instance, Spot, and capacity reservation data across your accounts and organization, providing actionable insights to optimize your AWS resources.

Images

Amazon Machine Images (AMIs)

An AMI contains the software configuration (for example, an operating system, an application server, and applications) required for the creation of EC2 instances. When you launch an instance, you must specify an AMI.

AMI Catalog

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. You can select an AMI provided by AWS, our user community, or the AWS Marketplace; or you can select one of your own AMIs.

Elastic Block Store

Volumes

Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

After you attach the volume to an instance, you can use it as you would use a physical hard drive.

Amazon EBS provides the following volume types:

- General Purpose SSD (gp3)
- General Purpose SSD (gp2)
- Provisioned IOPS SSD (io1)
- Provisioned IOPS SSD (io2)
- Throughput Optimized HDD (st1)
- Cold HDD (sc1)
- Magnetic (standard)

Snapshot

Create a point-in-time snapshot of an EBS volume and use it as a baseline for new volumes or for data backup. You can create snapshots from an individual volume, or you can create multi-volume snapshots from all of the volumes attached to an instance.

Lifecycle Manager

Amazon Data Lifecycle Manager provides a simple, automated way to back up data stored on Amazon EBS volumes.

Network & Security

Security group

A security group acts as a virtual firewall that controls the traffic for one or more instances. When you launch an instance, you can specify one or more security groups.

Elastic IPs

An Elastic IP address is a static IPv4 address designed for dynamic cloud computing. An Elastic IP address is allocated to your AWS account and is yours until you release it.

Placement group

Placement groups are a mechanism used in Amazon Elastic Compute Cloud (EC2) for controlling the placement of instances in a cluster.

Key Pairs

EC2 key pair consists of a public key and a private key that is used to securely communicate with EC2 instances. The public key is stored on the EC2 instance and is used to encrypt messages that can only be decrypted by the corresponding private key, which is kept by the user who generated the key pair. The private key is used to decrypt messages that have been encrypted with the public key

Network Interfaces

An elastic network interface is a logical networking component in a VPC that represents a virtual network card.

Load Balancing**Load balancers**

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Trust Stores

Trust stores offload client authentication to the Application Load Balancer and ensure only trusted clients communicate with your back-end applications.

Auto Scaling

Auto Scaling groups are collections of Amazon EC2 instances that enable automatic scaling and fleet management features. These features help you maintain the health and availability of your applications.